

Ayush Mahakulkar

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Professional Summary

Software Engineer with strong experience in Modern C++ (11/14/17), Python automation, and Linux-based systems. Hands-on experience in automotive networking (CAN, Ethernet, VLAN, TSYNC), protocol validation, and high-performance C++ tools. Proven ability to design automation frameworks, parsers, backend services, and scalable data-processing pipelines. Seeking roles in C++, Linux, Backend, or Automation-driven systems.

Technical Skills

Programming Languages: C++, Python, Shell Scripting

C++ Expertise: OOP, STL, Smart Pointers, Templates, Lambdas, Multithreading, Async, RAII, Design Patterns

Systems & Backend: Linux, Socket Programming, Client-Server Architecture, CMake, PostgreSQL

Automation & AI: LLM-Based Automation, RAG (ChromaDB), Prompt Engineering, Robot Framework, XTR, CI/CD (GitHub Actions), Docker

Automotive & Networking: CAN, CAN FD, Ethernet, VLAN (802.1Q), CAN TSYNC, ETHSYNC, IEEE 1722, UDP (Unicast/Multicast)

Tools: Git, Wireshark, EB Tresos, Trace32, ANDi SDK

Fundamentals: Data Structures & Algorithms, System Design Basics, LeetCode Problem Solving

Professional Experience

KPIT Technologies Ltd.

Associate Software Engineer

Pune

Jan 2025 – Present

- Developed a high-performance **C++ latency parser** on Linux to compute end-to-end CAN-Ethernet network latency using multithreading and async processing.
- Implemented **client-server socket communication** to stream parsed metrics to UI dashboards.
- Automated **CAN ↔ Ethernet routing** and **VLAN (802.1Q)** feature validation using C++, Shell, and Linux.
- Built scalable C++ automation binaries using **CMake** for repeatable protocol validation.
- Designed Python-based **BUS load generators** to inject UDP and CAN traffic for peak-load testing using ANDi SDK.

AI-Assisted Automation Platform (C++, Python, Linux)

- Architected an AI-driven automation platform to generate Gherkin scenarios, test keywords, and automation scripts.
- Designed deterministic prompt templates to ensure consistent LLM outputs aligned with enterprise standards.
- Implemented **YAML-based execution workflows** for scalable and configurable automation pipelines.
- Integrated Robot Framework and XTR for enterprise-grade execution and reporting.
- Built a **RAG system using ChromaDB** to retrieve reusable libraries and domain context.
- Automated generation of **10k+ test scripts per week**, achieving over **90% coverage**.
- Built CI/CD pipelines using GitHub Actions, reducing manual testing effort by **60%**.

KPIT Technologies Ltd.

Software Engineering Intern

Pune

Dec 2023 – Jan 2025

- Completed Genesis training in **Modern C++ (14/17/20)**, STL, OOP, multithreading, async, and system-level programming.
- Built mini-systems using smart pointers, design patterns, and concurrent programming.
- Completed advanced training in Python scripting and automation utilities.

Education

B.Tech in Computer Science

MIT Art, Design & Technology University, Pune

2024

Additional Information

- Regularly solving DSA problems on LeetCode and CodeChef.
- Building MERN stack projects with Dockerized deployments and CI pipelines.
- Open to relocation and remote roles; committed to continuous learning and upskilling.